

The Ultimate Guide to Polynomials

Everything you need to know about degrees, zeroes & secret math hacks! ✨



The Polynomial Family Tree

Linear

Degree: 1

Standard Form:
 $ax + b$



Quadratic

Degree: 2

Standard Form:
 $ax^2 + bx + c$



Cubic

Degree: 3

Standard Form:
 $ax^3 + bx^2 + cx + d$



💡 Degree = The **HIGHEST** power of x in the equation!

Wait, what actually IS a 'Zero'?

Math textbooks say:

A real number k is a zero of polynomial $p(x)$ if $p(k) = 0$.

TRANSLATION:

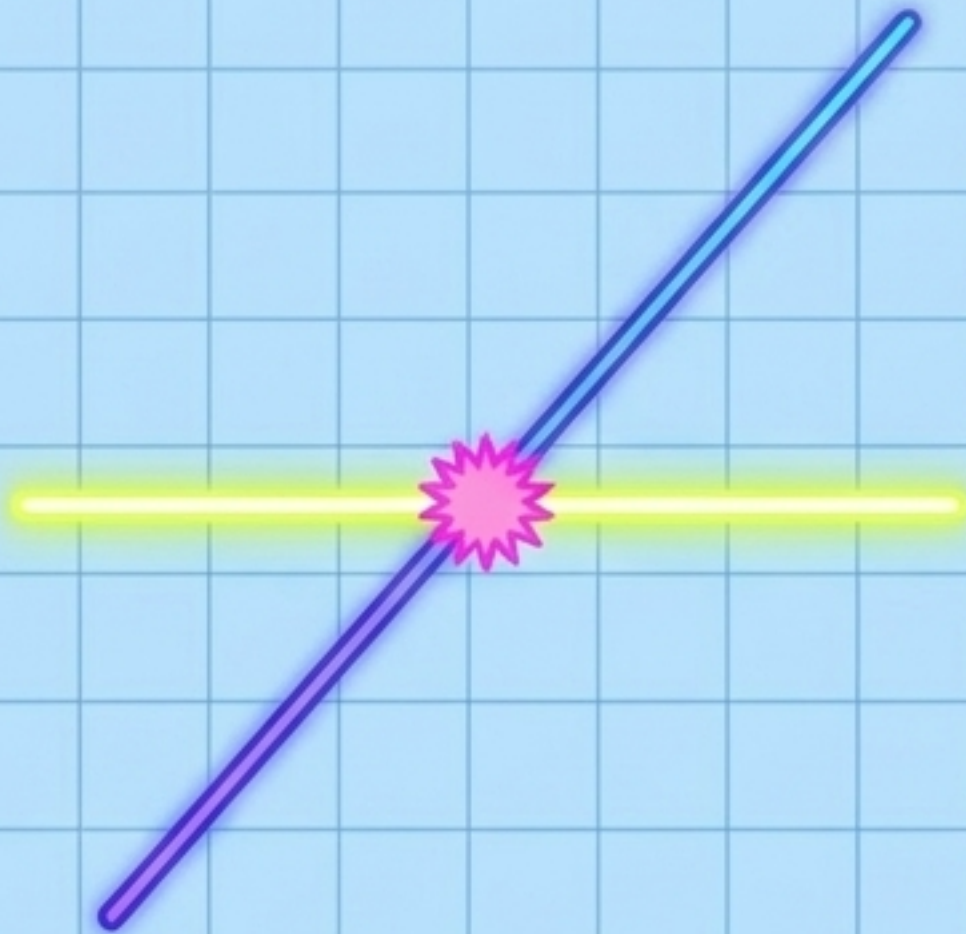
Real life says:

It's just the exact value of x that makes the whole equation equal ZERO!

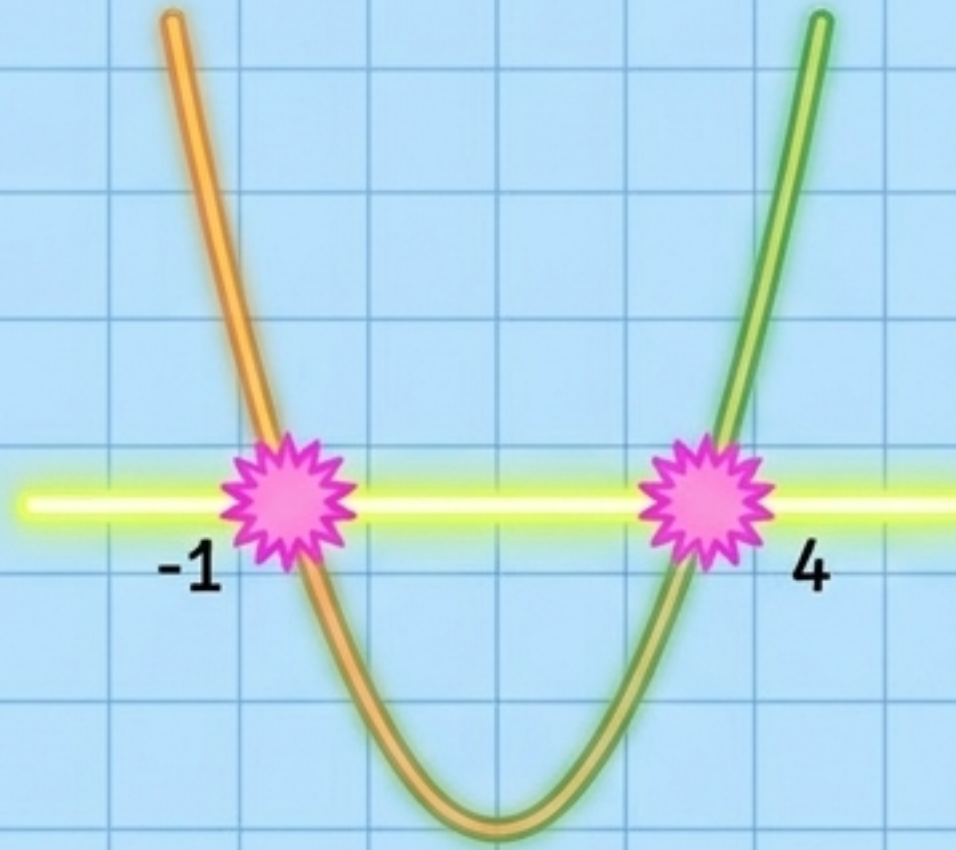
Example: $2x - 3$.
If $x = 3/2$, then
 $2(3/2) - 3 = 0$.
So, $3/2$ is the zero!



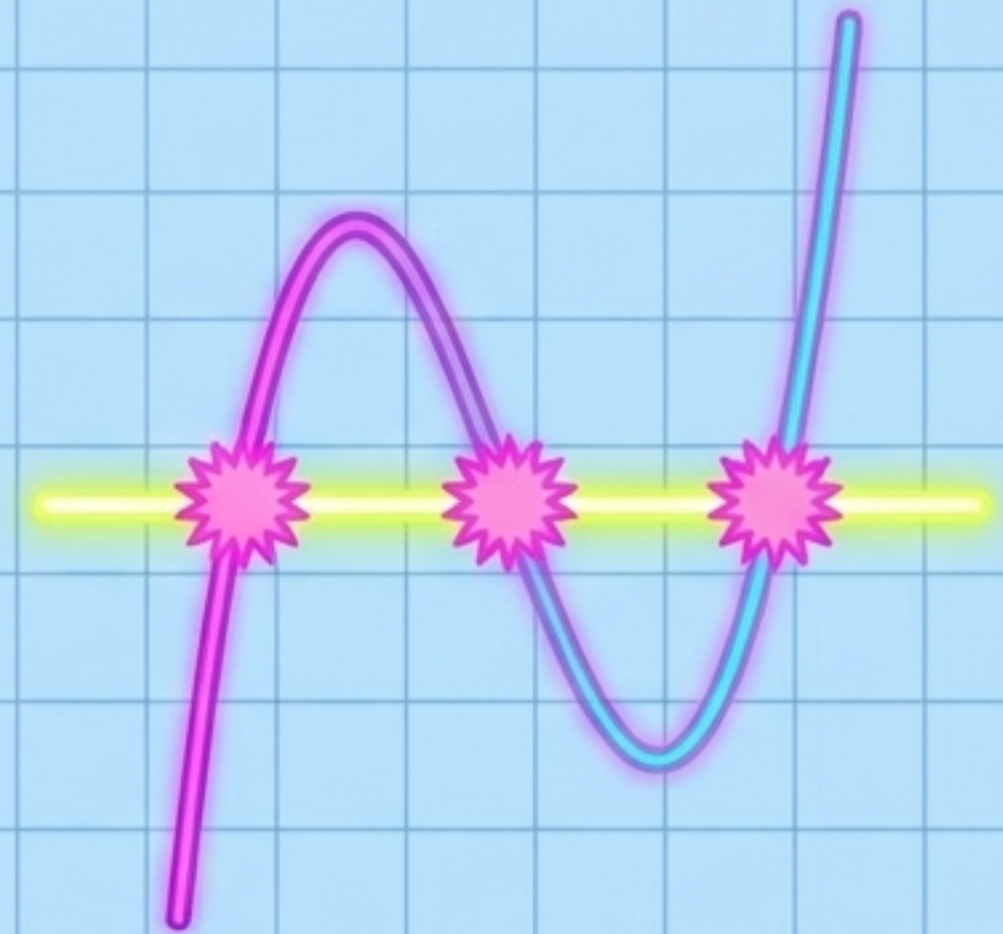
The Geometry of Zeros: Target Practice



Linear: 1 hit!



Quadratic: 2 hits!



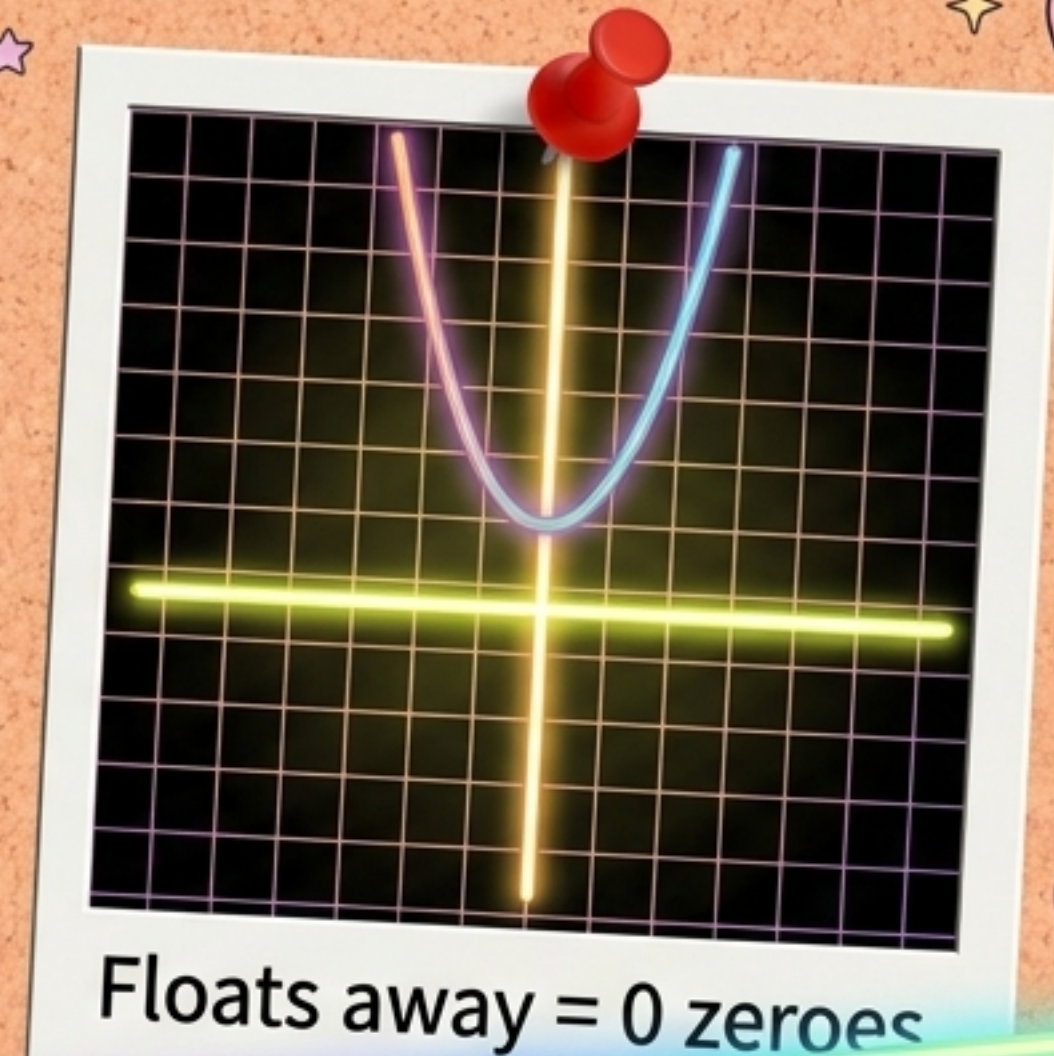
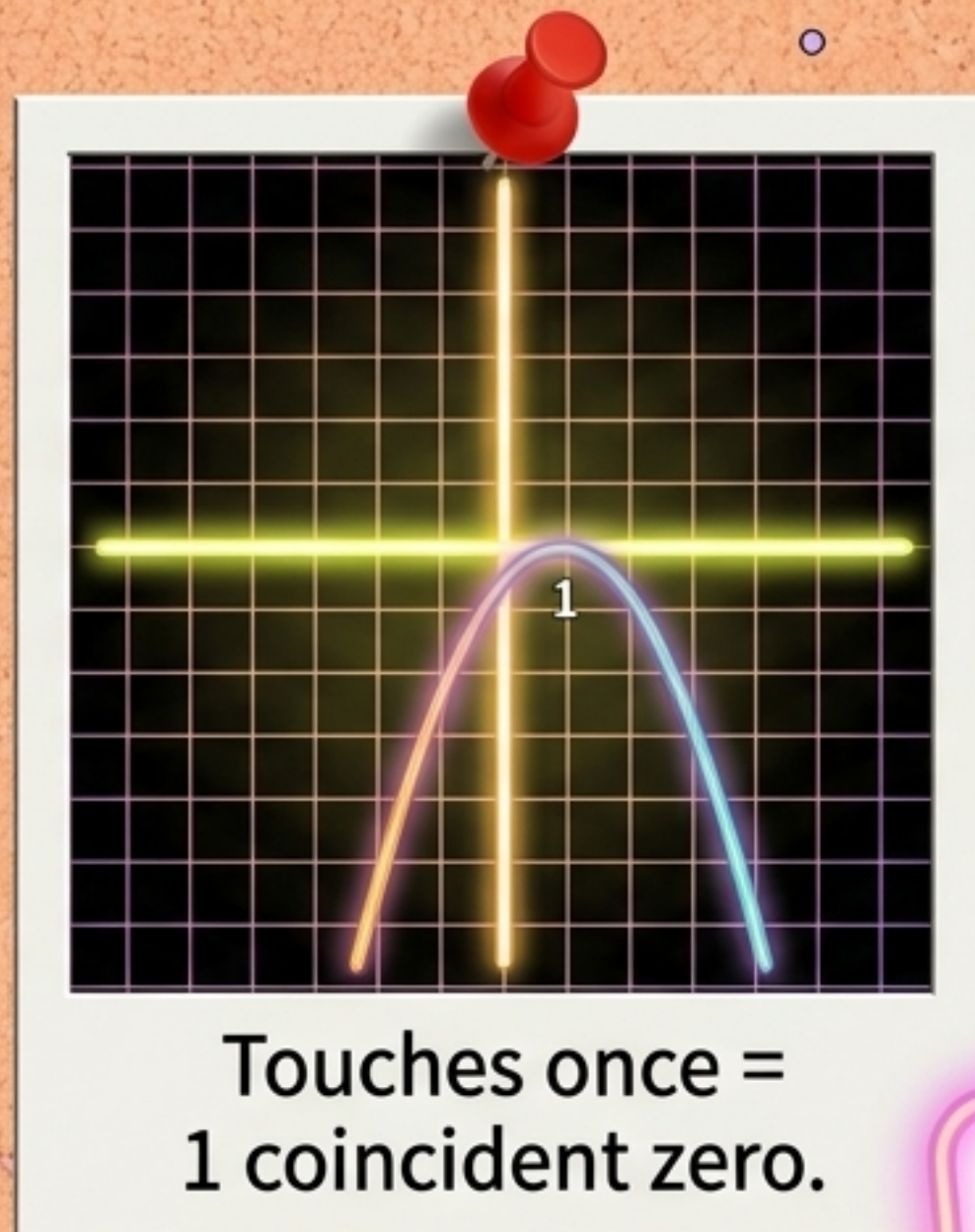
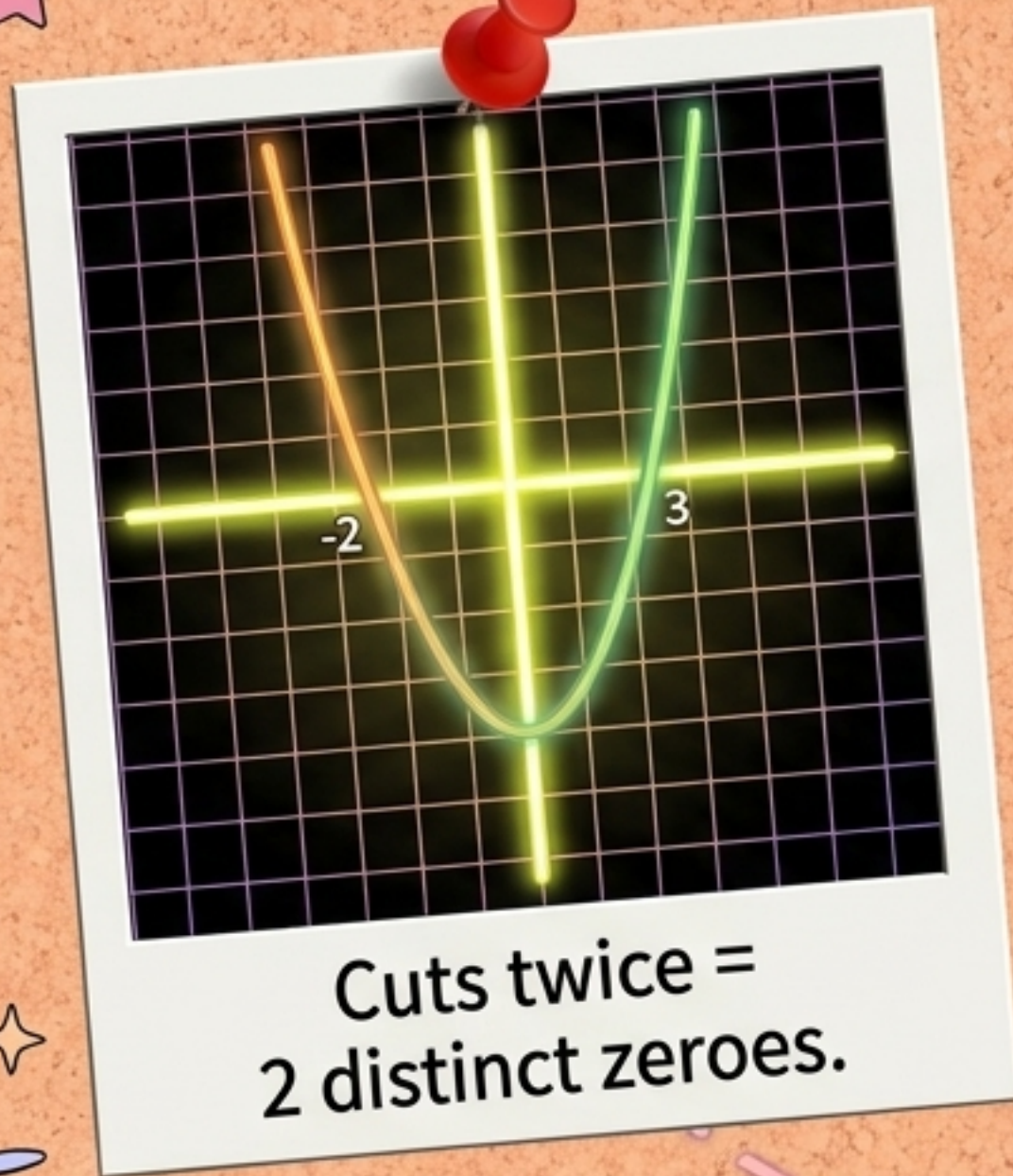
Cubic: 3 hits!



A 'Zero' is strictly the x-coordinate where the graph crosses the x-axis!



The Parabola Mood Board



★ **GOLDEN RULE:**
A polynomial of degree n
has a **MAXIMUM** of n zeroes!

The Secret Link: Quadratics



$$ax^2 + bx + c$$

Sum of zeroes
 $(\alpha + \beta) = -b/a$

Product of zeroes
 $(\alpha \times \beta) = c/a$

Let's test it on $x^2 + 7x + 10$!
- Sum = $-7/1 = -7$
- Product = $10/1 = 10$

The Final Boss: Cubics

$$ax^3 + bx^2 + cx + d$$

Sum:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

Sum of Products:

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$$

Product:

$$\alpha\beta\gamma = -\frac{d}{a}$$



Save this cheat sheet for your next exam!